Marketing Intelligence : Capstone Project - Athena Software
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1.
Decision 1 - Which game should Athena pursue?
Action Alternatives:

a. Acquire Warrior Guild
b. Acquire Seraph Guardians
c. Acquire Evercrest
d. Acquire no game.

Action Standards:

a. Maximise sales of the game within the first year of release.
b. Net revenue must exceed total acquisition costs (fixed + dev).
c. The game should be aligned with Athena's focus on premium Role Play Games.
d. The game should be relevant for Steam's market.

Decision 2 - How should the game be priced?
Action Alternatives:

a. Choose a launch price suggested in Gabor Granger responses for the selected game.

Action Standards:

a. Maximise first year net profit at that price (expected units at the price x price).
b. Ensure the price is consistent with the demand signal in the survey.

2.
a. <u>Market Size in 2019:</u>

Athena software specializes in premium role playing games for PC play. Using the 'Digital Games Revenue, 2019' chart from Superdata, the revenue generated by the Premium PC games is **$5.2B.**
Since Steam holds up to 75% of the global market share for PC games, the market size in dollars (market value) for the type of games Athena sells in 2019 would be
5.2 x 0.75 = $3.9B

b. <u>Market Size in 2020 (Pre Covid)</u>

From Superdata, the predicted revenue generated in 2020 is **$5.3B**.
Which means that the market size for the type of games Athena sells in 2020 would be
5.3 x 0.75 = $3.975B.
The pre covid forecast shows that premium games would have their biggest revenues yet, because highly anticipated games will launch in the first half of 2020.

c. <u>Expected Impact of Covid in 2020</u>

Covid 19 would have had a positive impact on this market. Consumers were forced to stay in their homes and would have had opportunities to take up gaming as a hobby. This would lead to an increase in game purchases across all the customer segments. The actual 2020 market would have substantially exceeded its forecasted value.

3.

a. <u>Factor Analysis:</u>

Before conducting the factor analysis the following two tests were conducted to verify if the data was suitable. (i) The Bartlett's Test of Sphericity gave a p value of 0, confirming that the correlation matrix is significantly different from the identity matrix and (ii) the KMO test gave an overall MSA of 0.890, which is well above the threshold 0.6. Both of the results confirm that Factor analysis can be done on the data.

Through PCA it was learnt that **11 factors** existed and were extracted using sklearns FactorAnalysis. The factor loadings are given below:

Rotated Factor Loadings:

1 to 40 of 40 entries Filter

Index	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8	Factor 9	Factor 10	Factor 11
imp.challenge	0.4250717906457834	0.0074913214923918B2	0.0538973101533453	-0.14820046759678965	-0.1444818048263966	0.00013665494253458645	0.0678807638197245	0.02357312489262637	-0.0276691150336114165	-0.0017767107791142691	-0.07376560083259
imp.unlocks	-0.1030436364939447	-0.7930325447453306	0.0864986332964859	-0.1356892144276694	0.0702996708723389	0.16154231687271748	0.00092815316570755	0.235769399453812B	-0.0096764295444390088	-0.1605294246286275	0.014067067418557568
imp.customize	0.0037328721851944938	-0.0119607958639137 2	-0.0963939723036421	-0.0962456744203093	-0.1146030483844413	-0.0055726716248808 5	-0.2205590920852949B	-0.022566525391967792	0.7636568484236894	0.0520339358706890 4	0.0317145285326915
imp.difficulty	0.4183183645001664	-0.0035507066021801 7	0.0802514096892433	-0.1368869747319106	-0.1376988221961483	0.0128501737411023	0.0811192097674774	0.020595426551799097	-0.0319471926341548	0.0081097856831205645	-0.0837420528975 5
imp.characters	-0.0247775297899334 5	0.0740430697497797	-0.7786728361304938	0.0440224886631016 2	-0.0770088345895742	-0.0307773022413424	-0.3065354243379966	0.012827282877704 7	0.1027026072869715	-0.0441679720940576	0.0320339538374196
imp.storyline	-0.0306303791947761424	0.0799647958675992	-0.7609563013077581 7	0.0561505641671362 6	-0.0879182436996833	-0.0501813979113594 7	-0.3350841088274825 7	0.005234344587739949	0.1311923880902377	-0.0720968004852646	0.0457018955761566
imp.mastery	0.6699325836400975	0.0319754690130215	0.0374618881194618 2	-0.1136107240734851 9	-0.1675830423511590 6	-0.0171521271131679 2	0.0317739634548023	-0.012753446915643 73	-0.0077637093697862 5	-0.0118349498787832 2	-0.4192707565893197
imp.backstory	-0.2299009795395084	0.0680845321249821 4	-0.7864845398146113	0.0428883547912101 5	-0.0780102339886682	-0.0084502085148286 8	-0.3063171696059171	0.00021584204187585 72	0.1159641519104638 7	-0.0600866930778996 1	0.0443034881173187
imp.dominate	-0.0223356991081581	0.1202267549461885	0.0497774158605662 6	0.0413909381307734	-0.0686617172075292 9	-0.1541910010817754 7	0.0118822324979462	-0.1660634198581648	0.0578192610273846 95	0.0894461401313562	-0.0366503359250442
imp.completion	-0.1062177321769727	-0.7971107693301516	0.0536163050297533 5	-0.1207693584095806	0.0700435300631785	0.1406483538441401	0.0215880780651852 5	0.209009289329176 2	0.0031866506617882 57	-0.1481447955233617	-8.66409632154973 2e-05
imp.wealth	0.0505194409383237	-0.0751466812542707	-0.0606465523497067 06	-0.0476288672552327	-0.1261880387075621 8	-0.0553201193240128 45	-0.2044830492685005 8	0.255463574824894 3	0.1430668062907553	-0.0240273089516403 2	-0.0226337330107223 15
imp.fantasy	-0.0164122647546035 3	0.0074392152992241 02	-0.1357075952795 5	-0.0767686002834096 3	-0.1005435493829384	0.0023821038890012 91	-0.7821594507464344	0.011348840497656028	0.1379859758428188	-0.0213307945072033	0.0332610773391422 54
imp.items	0.0040243259277991	0.0121257293646956 7	-0.2884156775767476	-0.0663497483992562	-0.0859827192797285	0.0029917683283237	-0.7145608380899425	-0.0090477400078563235	0.1163844113530666 1	-0.0425902474838425 75	0.0218491540929207 8
imp.power	-0.0081582899414314 52	-0.0086632312933459 5	-0.1443716178927106 5	-0.0514897542991469 14	-0.1119583661895484	0.0077671161105774 1	-0.7712927934924 51	0.00094038703513727 88	0.1166609957157467	-0.0190701986901806	0.0163326733642641
imp.offbeat	0.1871813879203788	0.0429833438622461	-0.0562190852028248	-0.0662826438572233	-0.7898245135865375	-0.0138777054062696	-0.1962225955260074 3	0.003899258054674 07	0.1016893067170906	0.0400672722297183 7	-0.0847785417285092
imp.collect	-0.1264187480212102	-0.7936742297465395	0.0872238682850157 4	-0.1189378535153434	0.0572885155703024	0.1469076606223454	-0.0027948143879524 86	0.235396939003428 76	0.0083263565653386	-0.1553134408791 59	0.0683771381449843 55
enj.excitement	0.0243482086234791 7	0.1113412940782213 3	-0.0285617820482463 46	0.1694033423033442 8	-0.0224190407869679 2	-0.7433447922411419	-0.0003280137706994 71	-0.0368388965708365 3	0.0081340361279908 6	0.1228726579130832	-0.0107728669836912 4
enj.destruction	-0.1146013394464246	0.0953462876757022 8	-0.0127457229623968 19	0.7876735719892789	0.0695255967084186 2	-0.1056292787614442	0.0998191528792425 6	-0.021396112090023643	-0.0955251715041104	0.2497704610769887	0.0629593716709125
enj.others	0.0996333760594643 73	0.1721704472424206	-0.0284249309611528 4	0.0204069843066226	-0.0587161578136752	-0.0767890293282549	0.0051836460073516 18	-0.7006124738573 17	0.0436057855363287 8	0.3064529301824784	0.0106984643737676 21
enj.react	0.0059928712249919 52	0.1185570018468506	-0.0297943321738491 6	0.1127541354524297	-0.0414020482442357	-0.7185159665295139	-4.2974527565205455e-05	-0.0090077195077901 35	0.0572299300440744 65	0.1310520116253063	0.0007868932147885 51
enj.duels	0.0048729969983578 6	0.1301505604199758	0.0480052923465249	0.0842570061491699	-0.0551365186267386	-0.1430217903835107 8	0.0347807288957414 5	-0.224571663941835	0.0337470119179683	0.7514516894042528	0.0132435093845955 4
enj.strategy	0.8055443387341 9	0.0683967934250901	-0.0113510999469399205	-0.1125006966602197	-0.1495470284207395	-0.0180989462712063	-0.0126152829847881 1	-0.0101616859541365 8	0.0154521971126208	-0.0108338104373634 87	-0.0377664745145153
enj.roleplay	-0.0158654597925328	0.0184981629293666426	-0.1330254259994428	-0.0513165661042377 65	-0.1043137058664290 5	0.0064977317174260 1	-0.7809495896729214	0.027528151270440473	0.1258145383141 4	0.0162818989190221 95	0.0295937122729635 5
enj.competition	9.89187557100863e-05	0.1451413372389287	0.0631293948935528	0.0860603046701319 5	-0.0270699199813352 5	-0.1423483023193367 8	0.0318661321052077 3	-0.216223337102973 4	0.0396512327049382 8	0.7618680652197865	0.0272783971417732 83
enj.decisions	0.8248470387731432	0.0708234773371449	0.0030041855855054 6	-0.1075155383945889	-0.1401583290595715	-0.0375353721566788 5	-0.0341148840533806	-0.116311335778783 5	0.0113852471751622 15	0.0170281451097955 12	-0.0233872154259469
enj.common_goal	0.0352990285980452 5	0.1784807019934108	-0.0041575569107041215	0.0193725840326417 63	-0.0716568870233189 2	-0.0456043795056289	-0.0340696722177941	-0.781098713828589 6	0.0644593568419868 8	0.1610718744376466	0.0088582528401031 07
enj.planning	0.7944119920936964	0.0992972946255020 5	-0.0082944453023355 97	-0.0926313444629396	-0.1568804935032738	-0.0239906632232925	0.0032373567460892 06	0.012569030979274 2	0.0092089649898668 45	-0.0045978266277671 6	-0.0706013252570513 1
enj.immersion	0.0014839658115634344	0.0151027694936118 1	-0.1363713750834714	-0.0999125345578599	-0.0964485918219239	0.0238101935185358	-0.7790678208575143	0.024425397994757268	0.1248527818419663 27	-0.0054922866878721 53	0.0474614893557584
enj.helping	0.1653303414557228	0.1673608271951948	-0.0108105402741027 32	0.0114414079376159 8	-0.0414846324391813	-0.0397812227231427 5	-0.0112021932062695 8	-0.7869822910888303	0.0593058025309620 16	0.1887983735885974	-0.0147389037710324 7
enj.fast	0.0418021992576532	0.1112545190507866	-0.0095948706344201 46	0.1243132145921312	-0.0305419814344899 2	-0.7523695010786 1	0.0067670606891033 6	-0.0314345934838395	0.0466797063945288 96	0.1368809977393155	0.0166560610586389 27
enj.guns	-0.1360513344308137 5	0.0863276809100339	-0.0448365197791451 56	0.7874564955389907	0.0554010349706406	-0.0902211125440167	0.0605752141143346	-0.0292518784657554	-0.0783751643838612	0.0431745290421309 6	0.0775005329377054 7
enj.gore	-0.1102052429637895	0.0741634837504589	-0.0255697635395362 42	0.7938103235307878	0.0884339097733826	-0.1456401484438506	0.0838481809339076	-0.0247496348084589	-0.0810861143030808	0.0571573171603778	0.0544233825667881
enj.blow.up	-0.1307639792588808	0.0783418101532183	-0.0350024865978108	0.7964937907156904	0.0816144410265795 5	-0.1212018016254351	0.0712181294006303	-0.0446802724285784 2	-0.1061127570541636	0.0897053408740595	0.0507373077888531
freq.explore	0.1787913366423576	0.0395644029403546	-0.0409958190303934	-0.0811983723191988	-0.7722876053283453	-0.0222547126598721	-0.1363733917660487	-0.0255104853613274 12	0.0758026598182072	0.0298444333939966	-0.0611435952373384
freq.experiment	0.1767370526792202	0.0332843264435360 65	-0.0524977113757287	-0.0783185345457202 8	-0.7789763445661 6	-0.0280627506560636	-0.1235998131487165 4	-0.0245354138417737 85	0.0951342220917843	0.0353577524419549 6	-0.0521738022592934 95
freq.study	0.6621093042922241	0.0514244502783436 96	0.0403320775471056 16	-0.1108658534526803	-0.1834223842126957 7	0.0067584885280566 2	0.0292835193001895 7	-0.0104444782757744 3	-0.0103750673508021	-0.0161157905966769 98	-0.3829763860263524
freq.char.creation	-0.0177738691155104 82	0.0131744598614308 99	-0.1049804126539792	-0.1294061287015725	-0.1122423258614708 7	-0.0800553731840513	-0.1816730383537892	-0.0226345352847850 5	0.7299175049827422	0.0259772120198489 45	0.0021598298383065465
freq.stats	0.0360359427149198	-0.0809380572804615	-0.0485019392783054	-0.0608831882450808	-0.1409084602176842	-0.0912500734164415	-0.2078663545570004	0.23872061983917	0.1421645821642496	-0.0301819902614046	-0.0263155383779972
freq.customize	0.0206961608768389	0.0101738320195332 89	-0.0951991947277566	-0.1084395115946754	-0.0934328863849284	-0.0280852936894357 9	-0.2011036162974948	-0.0157950376380811 27	0.7392925485967 87	0.0616411272288066 65	0.0139224090534622 24
freq.test.world	0.1792980672330594	0.0814761228262628	-0.0776902856855910 1	-0.0863876189053463	-0.7706639159161421	-0.0458592762852685	-0.1266017365513199	-0.0193040650640017 85	0.0901215315832975	0.0736495938247755	-0.0688481751249704 5

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The 11 factors that influence the customer segments, which are as follows:

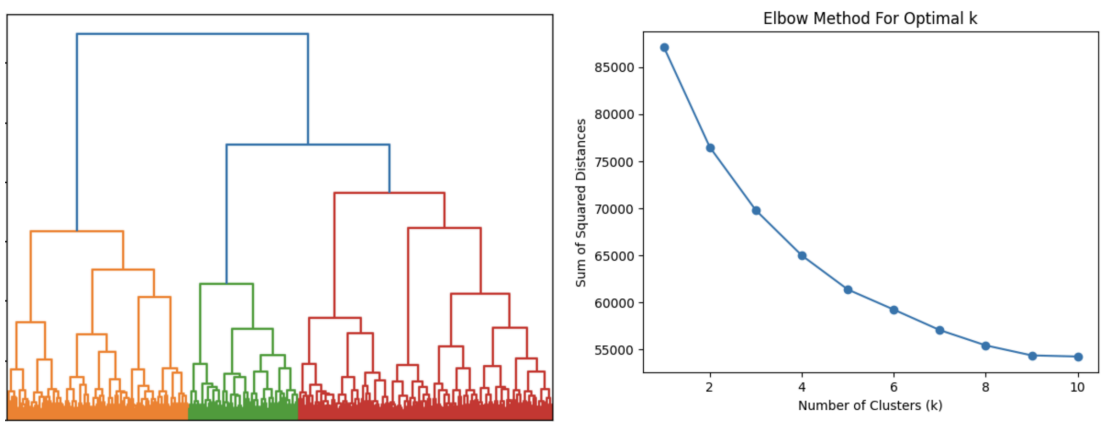
Factor 1: Strategy and Master Planning enj.decisions: 0.825 enj.strategy: 0.806 enj.planning: 0.794 imp.mastery: 0.670 freq.study: 0.662 imp.challenge: 0.425 imp.difficulty: 0.418	Factor 2: Anti-Completionism imp.completion: -0.797 imp.collect: -0.794 imp.unlocks: -0.759	Factor 3: Non-Narrative imp.backstory: -0.786 imp.characters: -0.779 imp.storyline: -0.761
Factor 4: Destruction Shooter Violence enj.guns: 0.797 enj.blow.up: 0.796 enj.gore: 0.794 enj.destruction: 0.788	Factor 5: Anti-Exploration imp.offbeat: -0.790 freq.experiment: -0.779 freq.explore: -0.772 freq.test.world: -0.771	Factor 6: Anti-Action enj.fast: -0.752 enj.excitement: -0.743 enj.react: -0.719
Factor 7: Non-Immersive imp.fantasy: -0.782	Factor 8: Solo Preference enj.helping: -0.787	Factor 9: Customization and Self Expression

enj.roleplay: -0.781 enj.immersion: -0.779 imp.power: -0.771 imp.items: -0.715	enj.common.goal: -0.781 enj.others: -0.701	imp.customize: 0.764 freq.customize: 0.739 freq.char.creation: 0.730
Factor 10: Competitive enj.competition: 0.762 enj.duels: 0.751 imp.dominate: 0.699	Factor 11: Anti-Challenge imp.difficulty: -0.684 imp.challenge: -0.677 imp.mastery: -0.419	

Several factors have negative loadings on their key variables, for these factors the factor names have been assigned to reflect the reversed direction, so a high score on any factor consistently indicates strong preference for what the factor is named.

b. <u>Cluster Analysis:</u>

To figure out the number of customer segments the market has, a dendrogram was built to decide how many segments to take. It suggested K = 3. To be sure about the analysis, the Elbow Method was also done which suggested a K=5. This reduces the variance within the clusters and captures finer-grained segment differences that are managerially meaningful for targeting decisions.



KMeans Clustering was performed and the cluster centers and summaries are below:

Cluster 0 : Strategic Completers
Size : 456 customers (20.9%)
Top Factors (Highest Scores):
- Factor 1: 1.22 - Strategy and Master Planning
- Factor 2: 0.43 - Anti-Completionism

Bottom Factors (Lowest Scores):
- Factor 4: -0.34 - Destruction Shooter Violence

Centers : (0.31823855 -0.61814163 0.66019012 0.27337926 0.60509319 0.5813828 0.49625664 0.59490661 0.42041331 -0.56849371 -0.06657694 0.52566378 0.62107696 0.57447036 0.79356366 -0.6190018 0.10749654 -0.35638093 0.73864882 0.16751754 0.47243512 0.51929676 0.56983145 0.4343697 0.55184218 0.8196985 0.53249225 0.52978419 0.73321944 0.17868869 -0.32832909 -0.31938058 -0.30453456 0.75526267 0.81693578 0.4933804 0.63405609 -0.04408599 0.65898075 0.81719812)

Cluster 1 : Casual Non-Strategists
Size : 543 customers (24.9%)
Top Factors (Highest Scores):
- Factor 2: 1.01 - Anti-Completionism
- Factor 4: 0.36 - Destruction Shooter Violence

Bottom Factors (Lowest Scores):
- Factor 1: -0.67 - Strategy and Master Planning
- Factor 3: -0.36 - Non-Narrative

Centers: (-0.57321233 -0.46703757 -0.35437972 -0.57999543 0.13889711 0.1453038 -0.61558497 0.13753103 0.33888495 -0.44456701 -0.22089464 -0.25546809 -0.2440529 -0.26317586 -0.42807144 -0.46623061 0.59129587 1.02944884 0.22835027 0.50042798 0.4046616 -0.563647 -0.23794314 0.43977789 -0.53349706 0.12603678 -0.53124456 -0.28319859 0.14708777 0.51539428 1.01930468 1.03263193 1.04988771 -0.42812226 -0.43378628 -0.61428132 -0.34620719 -0.22446505 -0.34261458 -0.37920666)

Cluster 2 : Passive Low-Engagement
Size : 345 customers (15.8%)
Bottom Factors (Lowest Scores):
- Factor 1: -1.12 - Strategy and Master Planning
- Factor 2: -0.81 - Anti-Completionism
- Factor 4: -0.71 - Destruction Shooter Violence

This suggests they value completion and dislike violence

Centers : (-0.55081719 0.98521422 0.00652725 -0.48073484 -0.62899882 -0.66888856 -0.73281311 -0.59891536 -0.35276628 0.98662238 -0.20921472 -0.36675454 -0.38956973 -0.34817963 -0.88045375 0.98831582 -0.61335993 -0.31562104 -0.33463801 -0.55453129 -0.33622885 -0.80799728 -0.32639732 -0.3387643 -0.83937166 -0.37592229 -0.85242549 -0.33386005 -0.33005951 -0.5710729 -0.28613718 -0.32448192 -0.297177 -0.85240785 -0.84111728 -0.74607282 -0.02419552 -0.21406918 -0.0275514 -0.87779368)

Cluster 3 : Narrative Avoiders
Size : 468 customers (21.5%)
Top Factors (Highest Scores):
- Factor 3: 1.36 - Non-Narrative

This suggests they are strongly indifferent to backstory, characters or storyline

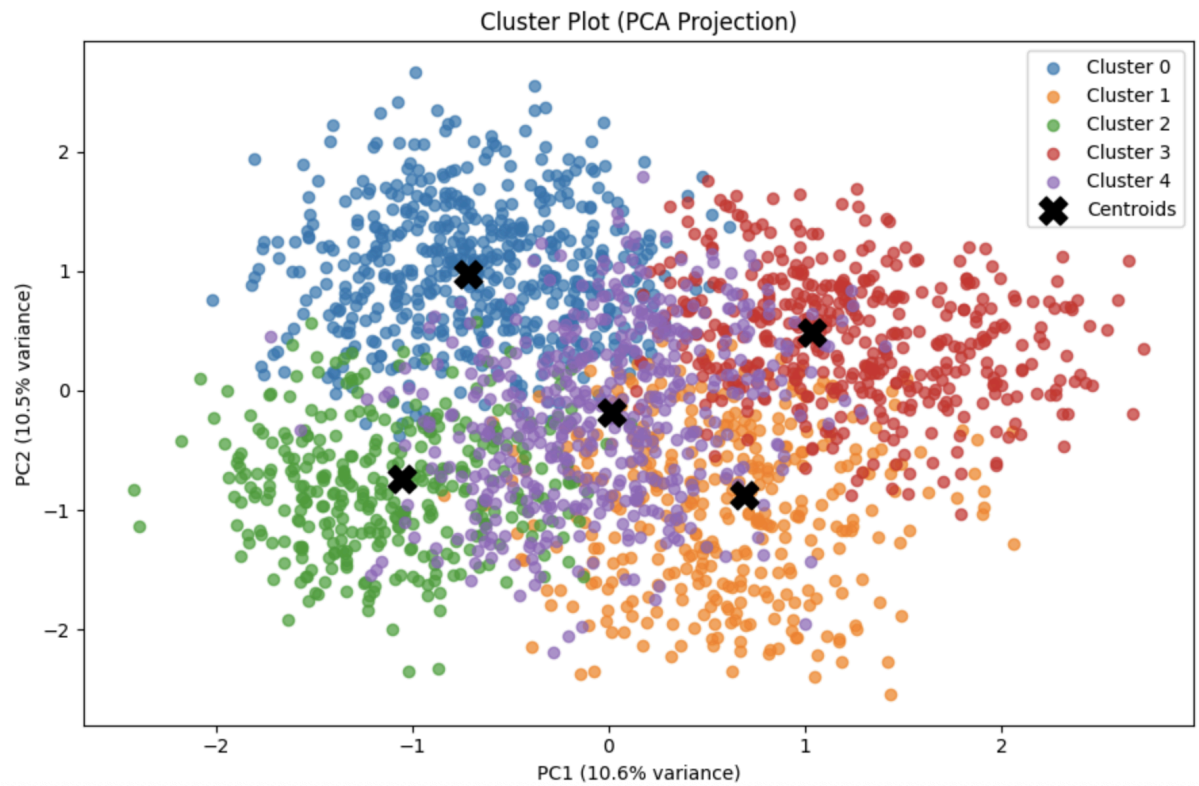
Centers : (0.84495928 -0.04459569 -0.62580805 0.85520452 -0.8255602 -0.81911592 0.8465207 -0.83786822 -0.11071247 -0.1011148 -0.18644648 -0.70407014 -0.75715985 -0.69557371 0.05279926 -0.08230329

-0.17254409 -0.23370528 -0.11113124 -0.17972672 -0.09499295 0.70040471 -0.72535185 -0.10751929 0.68580848 -0.05818043 0.70098715 -0.70257272 -0.04821617 -0.17780109 -0.29627817 -0.25050564 -0.29525521 0.0907029 0.04997257 0.8399306 -0.58895054 -0.20282059 -0.60323462 0.04400367)

Cluster 4 : Action Oriented Explorers
Size : 366 customers (16.8%)
Top Factors (Highest Scores):
- Factor 1: 0.60 - Strategy and Master Planning
- Factor 4: 0.35 - Destruction Shooter Violence

Bottom Factors (Lowest Scores):
- Factor 2: -0.98 - Anti-Completionism
- Factor 3: -0.89 - Non-Narrative

Centers : (-0.10729917 0.59138214 0.49728896 -0.12050715 0.68858783 0.73798438 -0.09667091 0.69068679 -0.55247321 0.5671312 0.84628663 0.97008643 0.92366666 0.8923401 0.40880944 0.63654925 -0.21238325 -0.48693356 -0.80152603 -0.1986202 -0.75056287 0.05527502 0.87822877 -0.73682859 0.01823422 -0.77950508 0.03189656 0.97317211 -0.75896454 -0.22161222 -0.45461439 -0.50791991 -0.52053489 0.38170045 0.35470364 -0.07409602 0.49955529 0.84907551 0.48459981 0.31560648)



c. <u>Demographic Profiles:</u>

Cross tabulation and regression analysis were used to examine the relationships between the cluster membership and demographics. The results are as below:

Cluster	Size	% Female	Average age	Average Income	Description
Strategic Completers	456	51.1%	22.9	$36568	Youngest and lower income segments
Casual Non-Strategists	543	35.5%	24.2	$41834	Mostly men, relatively young and modest income
Passive Low-Engagement	345	56.5%	28.6	$53823	Female skewed, middle range for age and income
Narrative Avoiders	468	42.1%	29.6	$59799	Male skewed, middle range for age higher income
Action Oriented Explorers	366	50%	36.3	$68940	Oldest and highest income

Gender is significantly associated with segment membership with a chi square statistic of 52.39 and $p<0.001$. However, Cramer's V = 0.11 indicates the association is weak in practical terms. Location is not significantly associated with segment membership. I performed a regression test for the states grouped into regions. The chi square statistic for region is 16.85 and $p = 0.155$ indicating segments are not meaningfully differentiated by geography. Regression analysis further shows that both age and income differ across segments significantly.

4.

a. <u>Handling Missing Data</u>

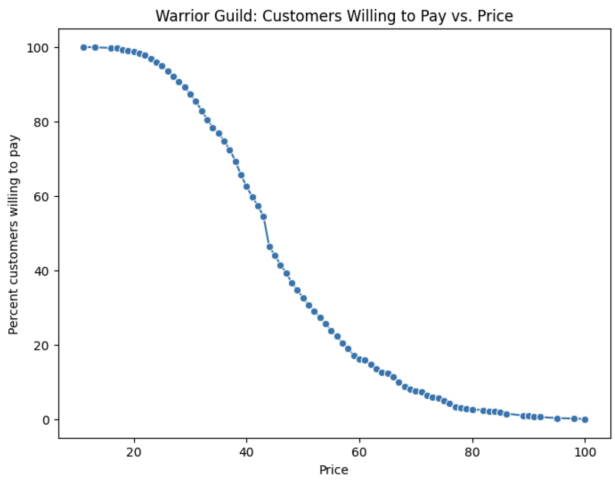
The only missing data in the Gabor Granger section was gg.maxprice, with 138 missing observations. Since the missingness rate was relatively small and was distributed similarly across the three games, I handled this by imputing the missing values using the median gg.maxprice within each presented game. I used game specific medians rather than a single overall value so that the price sensitivity pattern for each title was preserved. I also checked that results were directionally similar when using only complete cases.

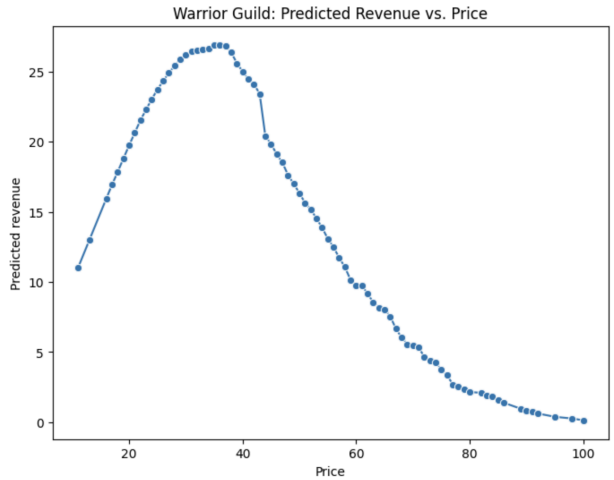
b. <u>Gabor Granger plots for each game:</u>

The ideal price is defined as the price that maximises predicted revenue.

Warrior Guild

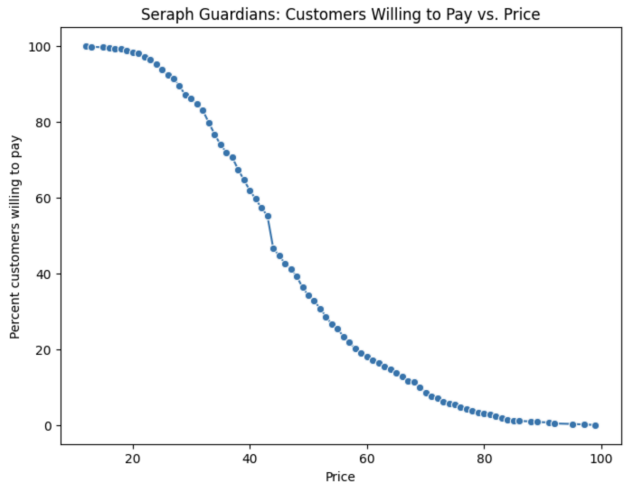
Willingness to pay = 76.88%
Ideal Price = $35.00

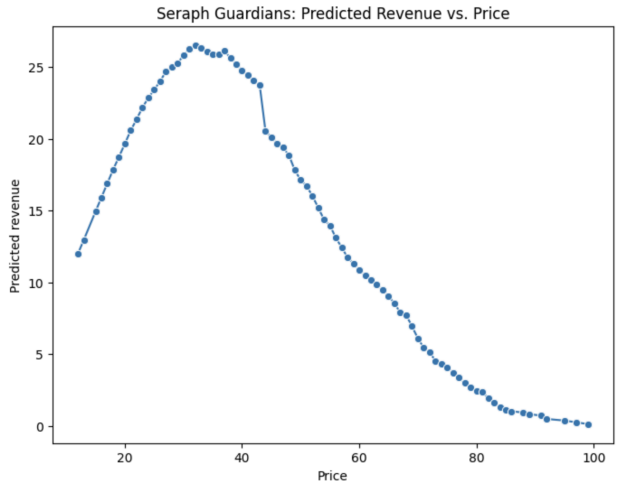




Seraph Guardians

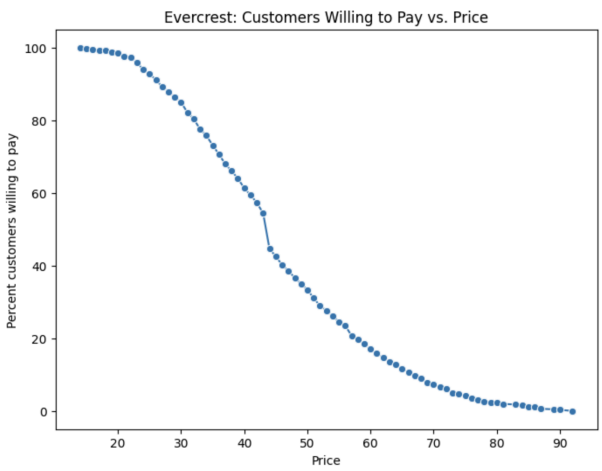
Willingness to pay = 82.94%
Ideal Price = $32.00

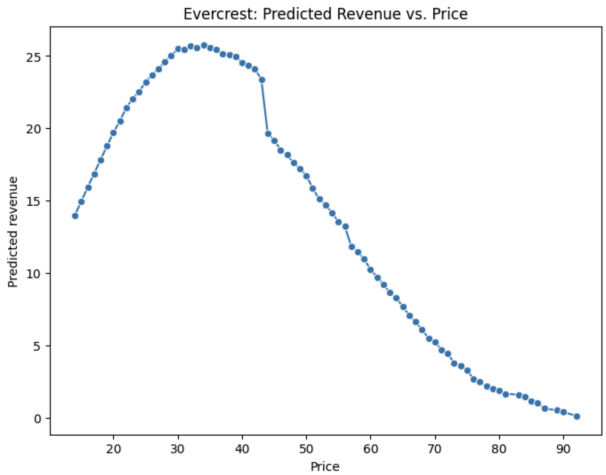




Evercrest

Willingness to pay = 75.81%
Ideal Price = $34.00





Seraph Guardians has the highest willingness to pay (82.94% as $32.00). Warrior Guild has the highest price people are willing to pay but at a very low willingness to pay percentage. Evercrest falls in between the two on both metrics.

b. Segment Interest By Game

The p-values for all the regressions performed are below 0.001.

Warrior Guild
The cluster most interested in the game is **Cluster 4 Action Oriented Explorers** and are willing to pay up to **$51.17**. The cluster least interested in the game is **Cluster 2 Passive Low-Engagement** who are willing to pay up to **$41.74**.

Seraph Guardians
The cluster most interested in the game is **Cluster 4 Action Oriented Explorers** and are willing to pay up to **$57.38**. The cluster least interested in the game is **Cluster 1 Casual Non-Strategists** who are willing to pay up to **$38.46**.

Evercrest
The cluster most interested in the game is **Cluster 4 Action Oriented Explorers** and are willing to pay up to **$52.94**. The cluster least interested in the game is **Cluster 1 Casual Non-Strategists** who are willing to pay up to **$37.87**.

This analysis suggests that the older and higher earning people are willing to pay more across all three games.

d. First Year Revenue Projections

Assuming 30% of respondents who indicated probable purchase actually convert within the first year, and that the survey sample is representative of the approximately 10 million active Steam customers interested in similar game types, the following first-year financial projections were

generated. Steam fees were applied using the tiered structure (30% up to $10M, 25% on $10M-$50M, 20% above $50M) and royalties were set at 5% of gross revenue.

Expected Buyers to Pay = (Total Customers) x (Willingness-to-Pay) x (conversion Rate) = 10 Million x (Willingness-to-Pay) x 0.3

Gross Revenue = (Ideal Price) x (Expected Buyers)

Steam Cut = steam_fee(Gross Revenue)

Royalty = 0.05 * Gross Revenue

Net Revenue = (Gross Revenue) - (Steam Cut) - Royalty

Profit = Net Revenue - Fixed Cost - Dev Costs

Metric	Warrior Guild	Seraph Guardians	Evercrest
Ideal Price	$35.00	$32.00	$34.00
% Willing to Pay	76.88%	82.94%	75.81%
Expected Buyers	2,306,452	2,488,095	2,274,336
Gross Revenue	$80,725,806	$79,619,048	$77,327,434
Steam Cut	$19,145,161	$18,923,810	$18,465,487
Royalties (5%)	$4,036,290	$3,980,952	$3,866,372
Net Revenue (pre-costs)	$57,544,355	$56,714,286	$54,995,575
Fixed + Dev Costs	$12,000,000	$12,500,000	$13,000,000
Profit After All Costs	$45,544,355	$44,214,286	$41,995,575

Warrior Guild yields the highest profit ($45.5M), followed by Seraph Guardians ($44.2M) and Evercrest ($42M). All three games are profitable and generate net revenue well in excess of their acquisition costs, confirming that any acquisition would create substantial value for Athena. Which erases the alternative of not acquiring any game.

5.
a. <u>Simulated Market Share by Action Alternative</u>

Assuming all games are priced equally and each customer purchases only their top-ranked game, market share under each action alternative was calculated in two possible scenarios.

1. Market share under each action alternative, by restricting the available games to only Athena's chosen game plus the three competitors
2. Market share under each action alternative was with all available games

These two scenarios were considered because in the case it was uncertain if the unacquired games will be acquired by another publisher in the same year or not.

Scenario 1:

Action Alternative	Athena Share	Top Competitor	Competitor Share
Acquire Warrior Guild	23.78%	Quest of the Titan	33.56%
Acquire Seraph Guardians	57.76%	Devil's Gate	32%
Acquire Evercrest	17.49%	Devil's Gate	36.5%

Scenario 2: since all games are released at once, only one market distribution exists for all action alternatives

Market dominated by	Athena Share	Top Competitor	Competitor Share
Seraph Guardians	50.05%	Devil's Gate	18.60%

Across all scenarios Seraph Guardians dominates the market share. If Athena acquires Seraph Guardians and is only competing against the confirmed competitors, it has a share of 57.76%. Whereas, the other two games don't have a powerful domination. In the second scenario as well Seraph Guardians looks like it'll perform exceedingly well. On a side note, Devil's Gate and Quest of Titans look like strong competitors.

b. <u>Assumptions and Limitations</u>

i. Equal Pricing

The simulation assumes that all six games are priced equally. However, the Gabor Granger analysis indicates different ideal prices for each Athena game. In reality price differences affect

purchase intent. This assumption likely inflates Seraph Guardians' estimated share slightly. Correcting this would reduce the estimated market share advantage of Seraph Guardians.

ii. One Purchase Per Customer

The model assumes that each customer purchases only one game. In reality a customer might buy many more games. This means that the total market opportunity is larger than assumed. A secondary purchase model, using ranks 2 and 3 to redistribute remaining share, would produce a more realistic picture of total revenue potential, particularly for games that rank highly but not first for many respondents.

6. Final Recommendations

Athena should acquire Seraph Guardians. This decision has been made based upon analysis on:

- Market Share : Seraph Guardians achieves 57.76% market share among surveyed Athena customers, higher than Warrior Guild or Evercrest. Even accounting for sample bias, this gap is substantial.
- Financial Performance: First-year profit of $44.2M is strong and only marginally below Warrior Guild, a difference unlikely to be meaningful given uncertainty in projections.
- Strategic Fit: Seraph Guardians is a single-player RPG with strategy and puzzle elements, which is directly aligned with Athena's existing catalog of premium PC RPGs. Warrior Guild, a multiplayer battle arena game, would represent a significant departure from Athena's brand identity.

Seraph Guardians needs to be priced at $32.00. This price is derived through the Gabor-Granger analysis with a willingness to pay at 82.94%.

It needs to be targeted towards the customers belonging to Cluster 4 Action Oriented Explorers and Cluster 0 Strategic Completers with a broad appealing message.

This targeted approach is used because these three segments showed the highest willingness to pay for this game. Their preference for games with strategy oriented gameplay that allows them to track their progress, makes them the most appropriate target audience for this title.

Positioning messaging should emphasize depth of strategy, richness of narrative, and the satisfaction of mastery. The multiple pathways to goal completion in Seraph Guardians directly appeals to both strategic planners and action-oriented explorers . A non-targeted approach is not recommended as the game's single-player, story-rich format is unlikely to appeal to Clusters 1 Casual Non-Strategists and has limited competitive advantage against multiplayer titles for those segments.

The pricing of the game is also not too bad because the targeted clusters' average age is 36 and they earn the highest income out of all the segments.